

Solution Code - Module 3-Core Java - FactorialCalculator

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q11_FactorialCalculator.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q11_FactorialCalculator {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.print("Enter a non-negative integer: ");
7 |
8 |         if (scanner.hasNextInt()) {
9 |             int number = scanner.nextInt();
10 |
11 |             if (number < 0) {
12 |                 System.out.println("Error: Factorial is not defined for negative numbers.");
13 |             } else {
14 |                 long factorial = 1; // using long to support values up to 20
15 |                 for (int i = 1; i <= number; i++) {
16 |                     factorial *= i;
17 |                 }
18 |                 System.out.println("Factorial of " + number + " is " + factorial + ".");
19 |             }
20 |         } else {
21 |             System.out.println("Error: Please enter a valid integer.");
22 |         }
23 |
24 |         scanner.close();
25 |     }
26 | }
```